United Nations

Our Common Future
Chapter 4: Population and Human Resources

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1. In 1985. some 80 million people were added to a world population of 4.8 billion. Each year the number of human beings increases, but the amount of natural resources with which to sustain this population, to improve the quality of human lives. and to eliminate mass poverty remains finite. On the other hand, expanding knowledge increases the productivity of resources.

2. Present rates of population growth cannot continue. They already compromise many governments' abilities to provide education, health care, and food security for people, much less their abilities to raise living standards. This gap between numbers and resources is all the more compelling because so much of the population growth is concentrated in low-income countries, ecologically disadvantaged regions, and poor households.

3. Yet the population issue is not solely about numbers. And poverty and resource degradation can exist on thinly populated lands, such as the drylands and the tropical forests. People are the ultimate resource. Improvements in education, health, and nutrition allow them to better use the resources they command, to stretch them further. In addition, threats to the sustainable use of resources come as much from inequalities in people's access to resources and from the ways in which they use them as from the sheer numbers of people. Thus concern over the 'population problem' also calls forth concern for human progress and human equality.

4. Nor are population growth rates the challenge solely of those nations with high rates of increase. An additional person in an industrial country consumer far more and places far greater pressure on natural resources than an additional person in the Third World. Consumption patterns and preferences are as important as numbers of consumers in the conservation of resources.

5. Thus many governments must work on several fronts to limit population growth; to control the impact of such growth on resources and, with increasing knowledge, enlarge their range and improve their productivity; to realize human potential so that people can better husband and use resources; and to provide people with forms of social security other than large numbers of children. The means of accomplishing these goals will vary from country to country, but all should keep in mind that sustainable economic growth and equitable access to resources are two of the more certain routes towards lower fertility rates.

6. Giving people the means to choose the size of their families is not just a method of keeping population in balance with resources; it is a way of assuring - especially for women the basic human right of self-determination. The extent to which facilities for exercising such choices are made available is itself a measure of a nation's development. In the same way. enhancing human potential not only promotes development but helps to ensure the right of all to a full and dignified life.

I. The Links with Environment and Development

7. Population growth and development are linked in complex ways. Economic development generates resources that can be used to improve education and health. These improvements, along with associated social changes, reduce both fertility and mortality rates. On the other hand, high rates of population growth that eat into surpluses available for economic and social development can hinder improvements in education and health.

8. In the past, the intensification of agriculture and the production of higher yields helped nations cope with the increasing population pressures on available land. Migration and international trade in food and fuels eased the pressure on local resources. They permitted and helped sustain the high population densities of some industrialized countries.

9. The situation is different in most of the developing world. There, improvements in medicine and public health have led to a sharp drop in mortality rates and have accelerated population growth rates to unprecedented levels. But fertility rates remain high; much human potential remains unrealized, and economic development is stalled. Agricultural intensification can go some way towards restoring a balance between food production and population, but there are limits beyond which intensification cannot go. (See Box 4-1.)

10. The very possibility of development can be compromised by high population growth rates. Moreover, most developing countries do not have the resources to wait for a few generations before population stabilizes. The option of migration to new lands is virtually closed. And low levels of economic and social development

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combined with changing trade production relationships limit possibilities of using international trade to augment access to resources. Hence, in the absence of deliberate measures, the imbalance between population growth and resource development will worsen.

11. Population pressure is already forcing traditional farmers to work harder, often on shrinking fame on marginal land, just to maintain household income. In Africa and Asia, rural population nearly doubled between 1950 and 1985, with a corresponding decline in land availability./1 Rapid population growth also creates urban economic and social problems that threaten to make cities wholly unmanageable. (See Chapter 9.)

12. Larger investments will be needed just to maintain the current inadequate levels of access to education, health care, and other services. In many cases, the resources required are just not available. Health, housing conditions, and the quality of education and public services all deteriorate; unemployment, urban drift, and social unrest increase.

Box 4-1
The Food/Population Balance

- The potential population-supporting capacity of land in developing countries has been assessed in a joint study by FAO and the International Institute for Applied Systems Analysis. Data on soil and land characteristics were combined with climatic data to calculate the potential yields of major crops, to select the optimum crops, and to derive the overall potential for calorie production. Three levels of crop production were calculated: the first at a low level of technology with no fertilizer or chemicals, traditional crop varieties, and no soil conservation; the second at an intermediate level, where the most productive crop mix is used on half the land along with fertilizers, improved varieties, and some soil conservation; and the third at a high level of technology with an ideal crop mix and technology on all lands. The population-supporting capacity was determined by dividing the total calorie production by a minimum per capita intake level. This figure was then compared with the medium-variant UN population projections.

- The 117 developing countries covered in the study, taken together, can produce enough food to feed one-and-a-half times their projected population in the year 2000. even at a low level of technology. But the picture is less hopeful in the cases of individual countries. At the low level of technology, 64 countries with a population of around 1.1 billion lack the resources to feed themselves. With the most advanced agricultural methods, the number of countries where food production potential would fall short of requirements drops to 19, with a total population of 100 million. Most are high-income West Asian countries and some small island states. Many of these countries have the capacity to earn enough foreign exchange to import their food requirements. In the others, the real issue is the modernization of agriculture on a sustainable basis.

- Some researchers have assessed the 'theoretical' potential for global food production. One study assumes that the area under food production can be around 1.5 billion hectares (close to the current level) and that average yields could go up to 5 tons of grain equivalent per hectare (as against the present average of 2 tons of grain equivalent}. Allowing for production from rangelands and marine sources, the total 'potential' is placed at 8 billion tons of grain equivalent.

- How many people can this sustain? The present global average consumption of plant energy for food, seed, and animal feed amounts to about 6,000 calories daily, with a range among countries of 3,000-15,000 calories, depending on the level of meat consumption. On this basis, the potential production could sustain a little more than 11 billion people. But if the average consumption rises substantially - say, to 9,000 calories - the population carrying capacity of the Earth comes down to 7.5 billion. These figures could be substantially higher if the area under food production and the productivity of 3 billion hectares of permanent pastures ran be increased on a sustainable basis. Nevertheless, the data do suggest, that meeting the food needs of an ultimate world population of around 10 billion would require tone changes in food habits, as well as greatly improving the efficiency of traditional agriculture.

B. Gilland. 'Considerations on World Population and Food Supply'. Population and Development Review. Vol. 9. No. 2. pp. 203-11; G M. Higgins et al., Potential Population supporting Capacities of Lands in the Developing

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World (Rome: FAO. 1982); D.J. Maler (ed.). Rapid Population Growth and Human Carrying Capacity. Staff Working Papers No. 690 (Washington, D.C.: World Bank, 1985).

13. Industrial countries seriously concerned with high population growth rates in other parts of the world have obligations beyond simply supplying aid packages of family planning hardware. Economic development, through its indirect impact on social and cultural factors, lowers fertility rates. International policies that interfere with economic development thus interfere with a developing nation's ability to manage its population growth. A concern for population growth must therefore be a part of a broader concern for a more rapid rate of economic and social development in the developing countries.

14. In the final analysis, and in both the developed and developing worlds, the population issue is about humane and not about numbers. It is misleading and an injustice to the human condition to see people merely fit; consumers. Their well being and security old age security, declining child mortality, health care, and so on are the goal o(development Almost any activity that increases well-being and security lessens peoples' desires to have more children than they and national ecosystems can support.

II. The Population Perspective

1. Growth in Numbers

15. Population growth accelerated in the middle of the 18th century with the advent of the Industrial Revolution and associated improvements in agriculture, not just in the regions that are more developed but elsewhere as well. The recent phase of deceleration started around 1950 with the sharp reduction in mortality rates in the developing countries.

Since 1970 it has been fashionable to draw a distinction between population and environment as two crisis areas, but often times we forget that population is in fact a very integral part of the environment and therefore when we are addressing ourselves to population we are looking at not only the physical, biological, and chemical environments, we are also looking at the socio-cultural or socio-economic environment in which these development programmes are being set. And population makes much more sense if you are talking of population within a context.

Dr. J.O. Oucho
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WCED Public Hearing
Nairobi. 23 Sept 1986

16. Between 1950 and 1985. world population grew at an annual rate of 1.9 per cent, compared with 0.8 per cent in the half-century preceding 1950./2 population growth is now concentrated in the developing regions of Asia, Africa, and Latin America, which accounted for 85 per cent of the increase of global population since 1950. (See Table 4-1.)

17. The processes of population growth are changing in most developing countries as birth and death rates fall. In the early 1950s, practically all developing countries had birth rates over 40 and death rates over 20, the major exception being the low death rates in Latin America. (These rates refer to the annual number of births and deaths per 1,000 population.) Today the situation is quite different:

- Thirty-two per cent of the people in the Third World live in countries - such as China and the Republic of Korea - with birth rates below 25 and death rates below 10.

- Forty-one per cent are in countries where birth rates have fallen, but not as much as death rates, and their populations are growing at around 2 per cent - doubling, in other words, every 34 years. Such countries include Brazil, India, Indonesia, and Mexico.

- The remaining 27 per cent live in countries, such as Algeria, Bangladesh. Iran, and Nigeria, where death rates have fallen slightly but birth rates remain high. Overall population growth is in the range of 2.5 to 3 per cent (doubling every 28 to 23 years), with even higher growth rates in some countries, such as Kenya./3

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18. In the industrial world, fertility rates have declined and the population is not growing rapidly. In fact, it has stabilized in many countries. Still, the population in North America, Europe, the USSR, and Oceania is expected to increase by 230 million by the year 2025. which is as many people as live in the United States today.

Table 4-1
World Population 1950-85: Key Facts

Size and Rates

	1950	1960	1970	1980	1985
Total Population	(billions)				
World	2.5	3.0	3.7	4.4	4.8
More developed regions	0.83	0.94	1.05	1.14	1.17
Less developed regions	1.68	2.07	2.65	3.31	3.66
Annual Growth *	(per cent)				
World		1.8	2.0	1.9	1.7
More developed regions		1.3	1.0	0.8	0.6
Less developed regions		2.1	2.5	2.3	2.0
Urban Population	(per cent)				
World	29	34	37	40	41
More developed regions	54	67	67	70	72
Less developed regions	17	22	25	29	31

* Data are for growth over previous decade or, for last column, over previous five years.

Source: Department of International Economic and Social Affairs, World Population Prospects: Estimates and Projections as Assessed in 1984 (New York: UN, 1986)

19. The acceleration of population growth in the Third World and the decline in fertility levels in industrial countries are (changing age distribution patterns radically. In developing countries, the young predominate. In 1900, 39 per cent of developing country populations were younger than 15; the figure for industrialized countries was only 23 per cent./4 Yet in these countries, the proportion of the elderly is growing. Those 65 or older accounted for 11 per cent of the population in 1980; in developing countries, they represented only 4 per cent./5 Thus in the industrial world, relatively fewer people of working age will bear the burden of supporting relatively larger numbers of older people.

20. A changing age structure helps to set patterns of future population growth. The large number of young people in developing countries means large numbers of future parents, so that even if each person produces fewer children. The total number of births will continue to increase. Population growth can continue to grow for some decades after fertility rates decline to the 'replacement level' of slightly over two children on average per couple. Thus in many nations, high population growth over the next few generations are assured.

21. Population projections indicate an increase in global population from 4.8 billion in 1985 to 6.1 billion by 2000, and to 8.2 billion by 2025. (See Table 4 2.) More than 90 pet cent of this increase is expected in developing regions. Large differences exist among countries in these areas, and the momentum of population growth is higher in Africa than in Latin America or Asia. In some developing countries, such as China, population growth rates are already well below 2 per cent and are expected to fall below 1 per cent by the beginning of the next century./6

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22. Reflecting the 'momentum' of population growth, long term UN projections show that at the global level:

- if replacement-level fertility is reached in 2010, global population will stabilize at 7.7 billion by 2060;
- if this rate is reached in 2025, population will stabilize at 10.2 billion by 2095;
- if. however, the rate is reached only in 2065, global population in 2100 would be 14.2 billion./7

23. These projections show that the world has real choices. Policies to bring down fertility rates could make a difference of billions to the global population next century. The greater part of the differences between the three variants is accounted for by South Asia. Africa, and Latin America. Hence much depends on the effectiveness of population policies in these regions.

2. Changes in Mobility

24. The number of people in Europe, Japan. North America. and the Soviet Union quintupled between 1750 and 1950. and these regions' share in world population increased sharply over thin period./8 By the latter part of the 19th century, there was growing concern about population pressures in Europe. Migration to North America, Australia, and New Zealand helped to some extent. At its peak between 1881 and 1910, permanent emigration absorbed nearly 20 per rent of the increase in population in Europe./9

25. Today, however, migration in not a major factor in determining population distribution among countries. Between 1970 and 1980 permanent emigration as a percentage of population increase fell to 4 per cent in Europe and was only 2.5 per cent in Latin America. The corresponding percentages in Asia and Africa were very much lower./10 Thus the option of emigration to new lands has not been and will not be a significant element in relieving demographic pressures in developing countries In effect, this reduces the time available to bring population into balance with resources.

26. Within countries, populations are more mobile. Improved communications have enabled large movements of people. Sometimes as a natural response to the growth of economic opportunities in different places. Some governments have actively encouraged migration from densely to sparsely settled areas. A more recent phenomenon is the flight of 'ecological refugees' from areas of environmental degradation.

Table 4-2
Current and Projected Population Size and Growth Rates*

	Population			Annual Growth Rate		
				1950	1985	2000
	1985	2000	2025	to	to	to
				1985	2000	2025
Region	(billion)			(per cent)		
World	4.8	6.1	8.2	1.9	1.6	1.2
Africa	0.66	0.87	1.62	2.6	3.1	2.5
Latin America	0.41	0.55	0.78	2.6	2.0	1.4
Asia	2.82	3.55	4.54	2.1	1.6	1.0
North America	0.26	0.30	0.35	1.3	0.8	0.6
Europe	0.49	0.51	0.52	0.7	0.3	0.1
USSR	0.28	0.31	0.37	1.7	0.8	0.6
Oceania	0.02	0.03	0.04	1.9	1.4	0.9

* Medium-variant projections.

Source: Department of International Economic and Social Affairs, World Population Prospects; Estimates and Projections as Assessed in 1984 (New York: UN, 1986).

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27. Much of the movement is from countryside to city. (See Chapter 9.) In 1985. some 40 per cent of the world's population lived in cities; the magnitude of the urban drift can be seen in the fact that since 1950, the increase in urban population has been larger than the increase in rural population both in percentage and in absolute terms. This shift is most striking in developing countries, where the number of city-dwellers quadrupled during this period./11

3. Improved Health and Education

28. Improvements in the health and education of all, but especially of women and in conjunction with other social changes that raise the status of women, can have a profound effect in bringing down population growth rates. In an initial period, however, better health care means that more babies live to reproduce and that women reproduce over longer time spans.

Table 4-3
Health Indicators

	Life Expectancy at Birth		Infant Mortality Rates	
	1950-55	1980-85	1960-65	1980-85
Region	(years)		(deaths per 1,000 live births)	
World	49.9	64.6	117	81
Africa	37.5	49.7	157	114
Asia	41.2	57.9	133	87
South America	52.3	64.0	101	64
North America	64.4	71.1	43	27
Europe	65.3	73.2	37	16
USSR	61.7	70.9	32	25
Oceania	61.0	67.6	55	39

Source: WCED, based on data in World Resources Institute / International Institute for Environment and Development, World Resources 1986 (New York: Basic Books, 1986).

29. The 'health status' of a society is a complex concept that cannot be measured easily. Two widely available indicators that reflect at least some aspects of a given society's health are life expectancy and infant mortality rates. (See Table 4-3.) These statistics suggest that health has improved virtually everywhere; and, at least with regard to these two indicators, the gap between industrial and developing regions has narrowed.

30. Many factors can increase life expectancy and reduce mortality rates: two are worth emphasizing. First, although generally speaking national wealth buys national health, some relatively poor nations and areas, such as China, Sri Lanka, and the Indian state of Kerala, have achieved remarkable success in lowering infant mortality and improving health through increases in education, especially of women; the establishment of primary health clinics; and other health care programmes./12 Second, the principal reductions in mortality rates in the industrial world came about before the advent of modern drugs; they were due to improved nutrition, housing, and hygiene. The recent gains in developing countries have also been largely due to public health programmes, particularly for the control of communicable diseases.

31. Education is another key dimension of 'population quality'. The past few decades have seen a great expansion of educational facilities in virtually all countries. In terms of school enrolment, literacy rates, the growth in technical education, and the development of scientific skills, much progress has been achieved. (See Table 4-4.)

III. A Policy Framework

32. Excessive population growth diffuses the fruits of development over increasing numbers instead of improving living standards in many developing countries: a reduction of current growth rates is an imperative for sustainable development. The critical issues are the balance between population size and available resources and the rate of population growth in relation to the capacity of the economy to provide for the basic needs of the

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population, not just today but for generations. Such a long-term view is necessary because attitudes to fertility rarely change rapidly and because, even after fertility starts declining, past increases in population impart a momentum of growth as people reach child-bearing age. However a nation proceeds towards the goals of sustainable development and lower fertility levels, the two are intimately linked and mutually reinforcing.

33. Measures to influence population size cannot be effective in isolation from other environment/development issues. The number, density, movement, and growth rate of a population cannot be influenced in the short run if these efforts are being overwhelmed by adverse patterns of development in other areas. Population policies must have a broader focus than controlling numbers: Measures to improve the quality of human resources in terms of health, education, and social development are as important.

34. A first step may be for governments to abandon the false division between 'productive' or 'economic' expenditures and 'social' expenditures. Policymakers must realize that spending on population activities and on other efforts to raise human potential is crucial to a nation's economic and productive activities and to achieving sustainable human progress - the end for which a government exists.

1. Managing Population Growth

35. Progress in population policies is uneven. Some countries with serious population problems have comprehensive policies. Some go no further than the promotion of family planning. Some do not do even that.

36. A population policy should set out and pursue broad national demographic goals in relation to other socio-economic objectives. Social and cultural factors dominate all others in affecting fertility. The most important of these is the roles women play in the family, the economy, and the society at large Fertility rates fall as women's employment opportunities outside the home and farm, their access to education, and their age at marriage all rise. Hence policies meant to lower fertility rates not only must include economic incentives and disincentives, but must aim to improve the position of women in society. Such policies should essentially promote women’s rights.

Table 4-4
Male and Female Enrolment Ratios, by Region, 1960 and 1982

		Male		Female	
Region		1960	1982	1960	1982
World					
	First Level	92.2	101.3	7.1	87.3
	Second Level	31.3	53.3	23.1	42.5
Africa					
	First Level	56.2	89.2	32.0	72.1
	Second Level	7.3	29.6	2.9	19.5
Latin America and Caribbean					
	First Level	75.0	106.2	71.2	103.3
	Second Level	14.9	46.6	13.6	48.5
North America					
	First Level	117.4	119.7	116.4	119.9
	Second Level	69.4	85.4	71.4	86.6

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Asia					
	First Level	94.9	100.1	63.1	79.9
	Second Level	29.3	49.3	16.6	32.9
Europe and USSR					
	First Level	103.4	105.4	102.7	104.5
	Second Level	46.5	76.2	44.6	81.3
Oceania					
	First Level	102.2	102.9	100.7	98.9
	Second Level	53.8	71.1	58.8	72.0

Note; The figures are percentages of appropriate groups receiving a given level of education. As many older children are in primary school percentages can be over 100.

Source: WCED, based on data in UNESCO, 'A Summary Statistical Review of Education in the World, 1960-1982, Paris, July 1984.

37. Poverty breeds high rates of population growth: Families poor in income, employment, and social security need children first to work and later to sustain elderly parents. Measures to provide an adequate livelihood for poor households, to establish and enforce minimum-age child labour laws, and to provide publicly financed social security will all lower fertility rates. Improved public health and child nutrition programmes that bring down infant mortality rates - so parents do not need 'extra' children as insurance against child death - can also help to reduce fertility levels.

> The environment is the business of everybody, development is the business of everybody, life and living is the business of everybody. I think the solution will be found in encouraging mass environmental literacy so that there can be democratic and literate decisions, because if decisions are taken by a few without the incorporation of the opinion of the masses, the NGOs especially included, the likelihood is that the situations will not succeed. They will be imposed from above, the people will not respond positively to them, and the project is lost before it is launched.
>
> Joseph Ouma
> Dean of School of Environmental Studies, Moi University
> WCED Public Hearing
> Nairobi, 23 Sept 1986

38. All these programmes are effective in bringing down birth rates only when their benefits are shared by the majority. Societies that attempt to spread the benefits of economic growth to a wider segment of the population may do better at lowering birth rates than societies with both faster and higher levels of economic growth but a less even sharing of the benefits of that growth.

39. Thus developing-country population strategies must deal not only with the population variable as such but also with the underlying social and economic conditions of underdevelopment. They must be multifaceted campaigns: to strengthen social, cultural, and economic motivations for couples to have small families and. through family planning programmes, to provide to all who want them the education, technological means, and services required to control family size.

40. Family planning services in many developing countries suffer by being isolated from other programmes that reduce fertility and even from those that increase motivation to use such services. They remain separate both in design and implementation from such fertility-related programmes as nutrition, public health, mother and

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child care, and preschool education that take place in the same area and that are often funded by the same agency.

41. Such services must therefore be integrated with other efforts to improve access to health care and education. The clinical support needed for most modern contraceptive methods makes family planning services heavily dependent on the health system. Some governments have successfully combined population programme: with health, education, and rural development projects, and implemented them as part of major socio-economic programmes in villages or regions. This integration increases motivation, improves access, and raises the effectiveness of investments in family planning.

42. Only about 1.5 per cent of official development aid now goes for population assistance./13 Regrettably, some donor countries have cut back on their assistance for multilateral population programmes and so weakened them; this must be reversed.

43. Zimbabwe is one nation that has successfully integrated its family planning efforts not only with its rural health services but also with efforts to improve women's abilities to organize group activities and earn money through their own labour. The government's initial efforts were aimed less at limiting population growth than at assisting women to space births in the interests of mother and child health and at helping infertile women to bear children. But gradually families have begun to use the contraceptives made available for child spacing as a way to limit fertility. Zimbabwe now leads sub-Saharan Africa in the use of modern contraceptive methods./14

2. Managing Distribution and Mobility

44. Population distribution across a country's different regions is influenced by the geographical spread of economic activity and opportunity. Most countries are committed in theory to balancing regional development, but are rarely able to do this in practice. Governments able to spread employment opportunities throughout their nations and especially through their countrysides will thus limit the rapid and often uncontrolled growth of one or two cities. China's effort to support village-level industries in the countryside is perhaps the most ambitious of this sort of national programme.

45. Migration from countryside to city is not in itself a bad thing; it is part of the process of economic development and diversification. The issue is not so much the overall rural urban shift but the distribution of urban growth between large metropolitan cities and smaller urban settlements. (See Chapter 9.)

46. A commitment to rural development implies more attention to realizing the development potential of all regions, particularly those that are ecologically disadvantaged (See Chapter 6.) This would help reduce migration from these areas due to lack of opportunities. But governments should avoid going too far in the opposite direction, encouraging people to cove into sparsely populated areas such as tropical moist forests, where the land may not be able to provide sustainable livelihoods.

Demographic phenomena constitute the heart of the African Development problematique. They are the data that lead most analysts to project a continuing and deepening crisis in Africa. There is no doubt of the imperative and urgent need for a far reaching population policy to be adopted and vigorously implemented by African governments.

One issue of relevance that requires further research is the use of the tax system as a means for controlling population growth and discouraging rural-urban migration.

To slow down population growth, should families without children be given a tax incentive or tax break? Should a tax penalty be imposed for each child after a fixed number of children, considering that the tax system has not solved the population migration problem?

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WCED Public Hearing
Harare, 18 Sept 1986

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3. From Liability to Asset

47. When a population exceeds the carrying capacity of the available resources, it can become a liability in efforts to improve people's welfare. But talking of population just as numbers glosses over an important point: People are also a creative resource, and this creativity is an asset societies must tap. To nurture and enhance that asset, people's physical well-being must be improved through better nutrition, health care, and so on. And education must be provided to help them become more capable and creative, skilful, productive, and better able to deal with day-to-day problems. All this has to be achieved through access to and participation in the processes of sustainable development.

3.1 Improving Health

> We in Asia, I feel, want to have an equilibrium between the spiritual and material life. I noticed that you have tried to separate religion from the technological side of life. Is that not exactly, the mistake in the West in developing technology, without ethics, without religion? If that is the case, and we have the chance to develop a new direction, should we not advise the group on technology to pursue a different kind of technology which has as its base not only the rationality, but also the spiritual aspect? Is this a dream or is this something we cannot avoid?
>
> Speaker from the floor
> WCED Public Hearing
> Jakarta, 26 March 1985

48. Good health is the foundation of human welfare and productivity. Hence a broad-based health policy is essential for sustainable development. In the developing world, the critical problems of ill health are closely related to environmental conditions and development problems.

49. Malaria is the most important parasitic disease in the tropics, and its prevalence is closely related to wastewater disposal and drainage. Large dams and irrigation systems have led to sharp increases in the incidence of schistosomiasis (snail fever) in many areas. Inadequacies in water supply and sanitation are direct causes of other widespread and debilitating diseases such as diarrhoeas and various worm infestations.

50. Though much has been achieved in recent years, 1.7 billion people lack access to clean water, and 1.2 billion to adequate sanitation./15 Many diseases can be controlled not just through therapeutic interventions but also through improvements in rural water supply, sanitation, and health education. In this sense, they really require a developmental solution. In the developing world, the number of water taps nearby is a better indication of the health of a community than is the number of hospital beds.

51. Other examples of links between development, environmental conditions, and health include air pollution and the respiratory illnesses it brings, the impact of housing conditions on the spread of tuberculosis, the effects of carcinogens and toxic substances, and the exposure to hazards in the workplace and elsewhere.

52. Many health problems arise from the nutritional deficiencies that occur in virtually all developing countries, but most acutely in low-income areas. Most malnutrition is related to a shortage of calories or protein or both, but some diets also lack specific elements and compounds, such as iron and iodine. Health will be greatly improved in low-income areas by policies that lead to the production of more of the cheap foods the poor traditionally eat - coarse grains and root crops.

53. These health, nutrition, environment, and development links imply that health policy cannot be conceived of purely in terms of curative or preventive medicine, or even in terms of greater attention to public health. Integrated approaches are needed that reflect key health objectives in areas such as food production; water supply and sanitation: industrial policy, particularly with regard to safety and pollution; and the planning of human settlements. Beyond this, it is necessary to identify vulnerable groups and their health risks and to ensure that the socio-economic factors that underlie these risks are taken into account in other areas of development policy.

54. Hence. WHO's 'Health for All' strategy should be broadened far beyond the provision of medical workers and clinics, to cover health-related interventions in all development activities./16 Moreover, this broader approach must be reflected in institutional arrangements to coordinate all such activities effectively.

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55. Within the narrower area of health care, providing primary health care facilities and making sure that everyone has the opportunity to use them are appropriate starting points. Maternal and child health care are also particularly important. The critical elements here are relatively inexpensive and can have a profound impact on health and well-being. An organized system of trained birth attendants, protection against tetanus and other childbirth infections, and supplemental feeding can dramatically reduce maternal mortality. Similarly, low-cost programmes to assure immunization, teach and supply oral dehydration therapy against diarrhoeas, and encourage breast-feeding (which in turn can reduce fertility) can increase child survival rates dramatically.

56. Health care must be supplemented by effective health education. Some parts of the Third World may soon face growing numbers of the illnesses associated with life-styles in industrial nations - cancer and heart disease especially. Few developing nations can afford the expensive treatment required for the latter diseases, and should begin efforts now to educate their citizens on the dangers of smoking and of high-fat diets.

57. A rapid spread of acquired immune deficiency syndrome (AIDS) in both developed and developing nations could drastically alter all countries' health priorities. AIDS is threatening to kill millions of people and disrupt the economies of many countries. Governments should overcome any lingering shyness and rapidly educate their people about this syndrome and about the ways in which it is spread. International cooperation on research and the handling of the disease is essential.

58. Another major health problem with international ramifications is the increase in drug addiction. It is a problem closely linked to organized crime in the production of drugs, in large-scale international traffic in these drugs, and in the networks for distribution. It distorts the economy in many poor producing areas and destroys people the world over. International cooperation is essential in tackling this scourge. Some countries have to deploy considerable financial resources to halt the production and traffic in narcotics and to promote crop diversification and rehabilitation schemes in the producing areas, which are generally impoverished. To sustain their efforts, greater international assistance is essential

59. Most medical research focuses on pharmaceuticals, vaccines. and other technological interventions for disease management. Much of this research is directed at the diseases of industrialized countries, as their treatment accounts for a substantial part of the sales of pharmaceutical companies. More research is urgently needed on the environmentally related tropical diseases that are the major health problem in the Third World, This research should focus not merely on new medicines, but also on public health measures to control these diseases. Existing arrangements for international collaboration on tropical disease research should be greatly strengthened.

Education and communication are vitally important in order to impress each individual of his or her responsibility regarding the healthy future of the earth. The best way for students to recognize that their action can make a difference is to have projects organized by the school or community on which the students can work. Once convinced that they can help, people tend to change both their attitude and their behaviour. New attitudes towards the environment will be reflected in decisions at home and in corporate boardrooms around the world.

Bernice Goldsmith
Student, North Toronto Collegiate
WCED Public Hearing
Ottawa, 26-27 May 1986

3.2 Broadening Education

60. Human resource development demands knowledge and skills to help people improve their economic performance. Sustainable development requires changes in values and attitudes towards environment and development - indeed, towards society and work at home, on farms, and in factories. The world's religions could help provide direction and motivation in forming new values that would stress individual and joint responsibility towards the environment and towards nurturing harmony between humanity and environment.

61. Education should also be geared towards making people more capable of dealing with problems of overcrowding and excessive population densities, and better able to improve what could be called 'social carrying capacities'. This is essential to prevent ruptures in the social fabric, and schooling should enhance the

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levels of tolerance and empathy required for living in a crowded world. Improved health, lower fertility, and better nutrition will depend on greater literacy and social and civic responsibility. Education can induce all these, and can enhance a society's ability to overcome poverty, increase incomes, improve health and nutrition, and reduce family size.

62. The investment in education and the growth in school enrolment during the past few decades are signs of progress. Access to education is increasing and will continue to do so. Today almost all the world's boys are getting some form of primary education. In Asia and Africa, however, enrolment rates for girls are much lower than for boys at all levels. A large gap also exists between developed and developing countries in enrolment rates beyond primary schools, as Table 4-4 indicated.

63. UN projections of enrolment rates for the year 2000 suggest a continuation of these trends. Thus despite the growth in primary education, illiteracy will continue to rise in terms of sheer numbers: there will be more than 900 million people unable to read and write at the end of the century. By then, girls' enrolment rates are still expected to be below the current rates for boys in Asia. As for secondary education, developing countries are not expected to attain even the 1960 industrial country levels by the year 2000./17

64. Sustainable development requires that these trends be corrected. The main task of education policy must be to make literacy universal and to close the gaps between male and female enrolment rates. Realizing these goals would improve individual productivity and earnings, as well as personal attitudes to health, nutrition, and child-bearing. It can also instill a greater awareness of everyday environmental factors. Facilities for education beyond primary school must be expanded to improve skills necessary for pursuing sustainable development.

65. A major problem confronting many countries is the widespread unemployment and the unrest that it leads to. Education has often been unable to provide the skills needed for appropriate employment. This is evident in the large numbers of unemployed people who have been trained for white-collar employment in swelling urban populations. Education and training should also be directed towards the acquisition of practical and vocational skills, and particularly towards making people more self-reliant. All this should be supported by efforts to nurture the informal sector and the participation of community organizations.

66. Providing facilities is only the beginning. Education must be improved in quality and in relevance to local conditions. In many areas, it should be integrated with children's participation in farm work, a process requiring flexibility in the school system. It should impart knowledge relevant for the proper management of local resources. Rural schools must teach about local soils, water, and the conservation of both, about deforestation and how the community and the individual can reverse it. Teachers must be trained and the curriculum developed so that students learn about the agricultural balance sheet of an area.

67. Most people base their understanding of environmental processes and development on traditional beliefs or on information provided by a conventional education. Many thus remain ignorant about ways in which they could improve traditional production practices and better protect the natural resource base. Education should therefore provide comprehensive knowledge, encompassing and cutting across the social and natural sciences and the humanities, thus providing insights on the interaction between natural and human resources, between development and environment.

68. Environmental education should be included in and should run throughout the other disciplines of the formal education curriculum at all levels - to foster a sense of responsibility for the state of the environment and to teach students how to monitor, protect, and improve it. These objectives cannot be achieved without the involvement of students in the movement for a better environment, through such things as nature clubs and special interest groups. Adult education, on-the-job training, television, and other less formal methods must be used to reach out to as wide a group of individuals as possible, as environmental issues and knowledge systems now change radically in the space of a lifetime.

69. A critical point of intervention is during teacher training. The attitudes of teachers will be key in increasing understanding of the environment and its. links with development. To enhance the awareness and capabilities of teachers in this area, multilateral and bilateral agencies must provide support for the relevant curriculum development in teacher training institutions, for the preparation of teaching aids, and for other similar activities. Global awareness could be fostered by encouraging contacts among teachers from different countries, for instance in specialized centres set up for this purpose.

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I am here as the son of a small nation, the Krenak Indian Nation. We live in the valley of the Rio Doce. which is the frontier of Espirito Santo with the State of Minas Gerais. We are a micro-country - a micro-nation.

When the government took our land in the valley of Rio Doce, they wanted to give us another place somewhere else. But the State, the government will never understand that we do not have another place to go.

The only possible place for the Krenak people to live and to re-establish our existence, to speak to our Gods, to speak to our nature, to weave our lives is where our God created us. It is useless for the government to put us in a very beautiful place, in a very good place with a lot of hunting and a lot of fish. The Krenak people, we continue dying and we die insisting that there is only one place for us to live.

My heart does not become happy to see humanity's incapacity. I have no pleasure at all to come here and make these statements. We can no longer see the planet that we live upon as if it were a chess-board where people just move things around. We cannot consider the planet as something isolated from the cosmic.

We are not idiots to believe that there is possibility of life for us outside of where the origin of our life is. Respect our place of living, do not degrade our living condition, respect this life. We have no arms to cause pressure, the only thing we have is the right to cry for our dignity and the need to live in our land.

Ailton Krenak
Coordinator of Indian Nations Union
WCED Public Hearing
Sao Paulo. 28-29 Oct 1985

3.3 Empowering Vulnerable Groups

70. The processes of development generally lead to the gradual integration of local communities into a larger social and economic framework. But some communities - so-called indigenous or tribal peoples - remain isolated because of such factors as physical barriers to communication or marked differences in social and cultural practices. Such groups are found in North America, in Australia, in the Amazon Basin, in Central America, in the forests and hills of Asia, in the deserts of North Africa, and elsewhere.

71. The isolation of many such people has meant the preservation of a traditional way of life in close harmony with the natural environment. Their very survival has depended on their ecological awareness and adaptation. But their isolation has also meant that few of them have shared in national economic and social development; this may be reflected in their poor health, nutrition, and education.

72. With the gradual advance of organized development into remote regions, these groups are becoming less isolated. Many live in areas rich in valuable natural resources that planners and 'developers' want to exploit, and this exploitation disrupts the local environment so as to endanger traditional ways of life. The legal and institutional changes that accompany organized development add to such pressures.

73. Growing interaction with the larger world is increasing the vulnerability of these groups, since they are often left out of the processes of economic development. Social discrimination, cultural barriers, and the exclusion of these people from national political processes makes these groups vulnerable and subject to exploitation. Many groups become dispossessed and marginalized, and their traditional practices disappear. They become the victims of what could be described as cultural extinction.

74. These communities are the repositories of vast accumulations of traditional knowledge and experience that links humanity with its ancient origins. Their disappearance is a loss for the larger society, which could learn a great deal from their traditional skills in sustainably managing very complex ecological systems. It is a terrible irony that as formal development reaches more deeply into rain forests, deserts, and other isolated environments, it tends to destroy the only cultures that have proved able to thrive in these environments.

75. The starting point for a just and humane policy for such groups is the recognition and protection of their traditional rights to land and the other resources that sustain their way of life - rights they may define in terms that do not fit into standard legal systems. These groups' own institutions to regulate rights and obligations are crucial for maintaining the harmony with nature and the environmental awareness characteristic of the

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traditional way of life. Hence the recognition of traditional rights must go hand in hand with measures to protect the local institutions that enforce responsibility in resource use. And this recognition must also give local communities a decisive voice in the decisions about resource use in their area.

76. Protection of traditional rights should be accompanied by positive measures to enhance the well-being of the community in ways appropriate to the group's life-style. For example, earnings from traditional activities can be increased through the introduction of marketing arrangements that ensure a fair price for produce, but also through steps to conserve and enhance the resource base and increase resource productivity.

77. Those promoting policies that have an impact on the lives of an isolated, traditional people must tread a fine line between keeping them in artificial, perhaps unwanted isolation and wantonly destroying their life-styles. Hence broader measures of human resource development are essential. Health facilities must be provided to supplement and improve traditional practices; nutritional deficiencies have to be corrected, and educational institutions established. These steps should precede new projects that open up an area to economic development. Special efforts should also be made to ensure that the local community can derive the full benefit of such projects, particularly through jobs.

78. In terms of sheer numbers, these isolated, vulnerable groups are small. But their marginalization is a symptom of a style of development that tends to neglect both human and environmental considerations. Hence a more careful and sensitive consideration of their interests is a touchstone of a sustainable development policy.

Footnotes

1/ Department of International Economic and Social Affairs (DIESA). World Population Prospects; Estimates and Projections as Assessed in 1984 (New York: United Nations, 1986).

2/ Ibid.

3/ Based on data from UNCTAD, Handbook of International Trade and Development Statistics 1985 Supplement (New York: 1985).

4/ World Bank, World Development Report 1984 (New York: Oxford University Press, 1984).

5/ Ibid.

6/ DIESA. op. cit.

7/ UN. Population Bulletin of the United Nations. No. 14. 1982 (New York: 1983).

8/ C. Clark, Population Growth and Land Use (New York: St. Martin's Press. 1957).

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10/ Ibid.

11/ DIESA, op. cit.

12/ WHO, Intersectoral Linkages and Health Development, Case Studies in India (Kerala State), Jamaica. Norway. Sri Lanka and Thailand (Geneva: 1984).

13/ World Bank, op. cit.

14/ L. Timberlake. Only One Earth: Living for the Future (London: BBC/Earthscan. 1987).

15/ UNEP, The State of the Environment: Environment and Health (Nairobi: 1986).

16/ WHO. Global Strategy for Health for All by the Year 2000 (Geneva: 1981).

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17/ UNESCO. A Summary Statistical Review of Education in the World. 1960-82 (Paris: 1984).

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